



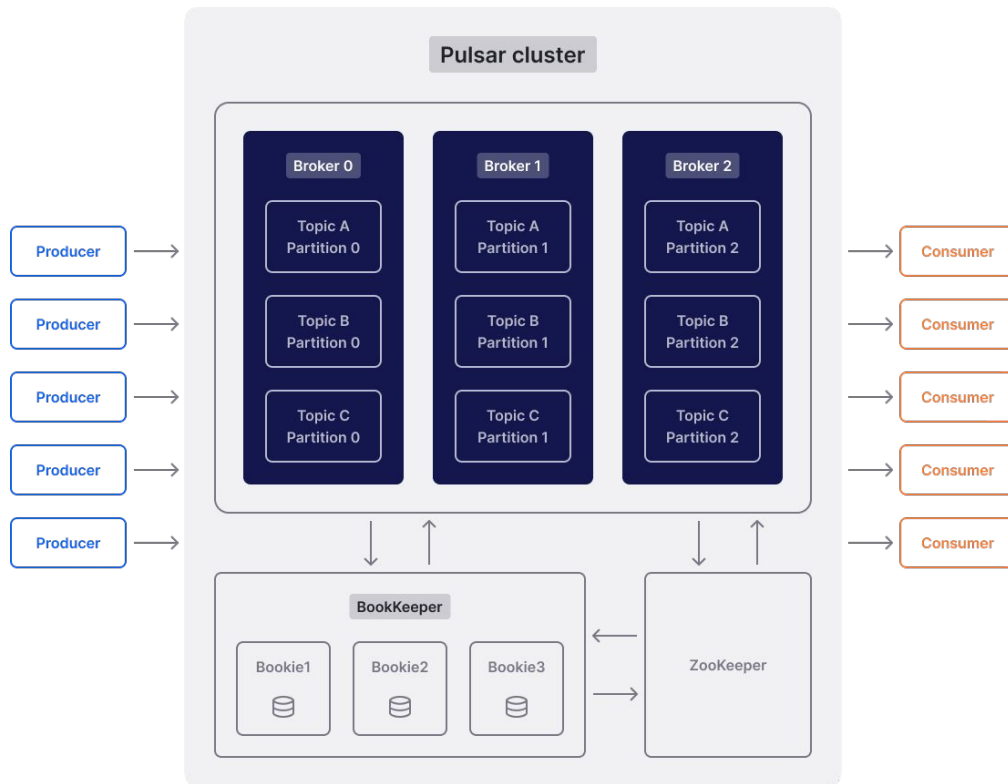
Apache Pulsar

What is it?



- A cloud-native, distributed messaging and streaming platform
- Originally created by Yahoo, open-source on 2016
- Support traditional message queuing and continuous event streaming
- Uses a decoupled architecture that separates compute (brokers) from storage (Apache BookKeeper)

Architecture



Trade-offs



Pros

- Decoupled Architecture
- Native multi-tenancy
- Built-in tiered-storage
- Unified Queue and Streaming models
- Built-in geo-replication
- Pulsar Functions
- Schema integrated
- Persistent and in memory topics



Cons

- Operational Complexity (Broker, Bookkeeper, ZooKeeper)
- Small ecosystem compared with Kafka
- Higher latency baseline
- Few cloud offers (StreamNative, IBM Astra Streaming,)

Benchmark



Do yourself.

Comparisons



Feature	Apache Kafka	Apache Pulsar
Programming Languages	17 supported	6 supported
Stack Overflow Questions	33,281 questions	358 questions
Connectors	400+ open source	~20 connectors
Multi-tenancy	Limited support	Native, built-in
Geo-replication	Requires additional tools	Built-in at topic/namespace levels
Schema Registry	Separate component	Built-in
Geo-Replication	Needs MirrorMaker	Built-in





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